What separates a fraudulent card transaction from a normal one? Evidence from 1.85 million simulated transactions

Group name	Null & Void
Dataset	kaggle.com/datasets/kartik2112/fraud-detection
Code	github.com/DarshanBothra/DSc-Project/tree/main/midterm (code in `midterm/src`, README in `midterm/README.md`)

#	Name	Roll no.	Institute email	Contribution
1	Darshan Bothra	2024176	darshan24176@iiitd.ac.in	Dataset choice, repository, classifier design (1, 5)
2	Piyush Keshan	2023376	piyush23376@iiitd.ac.in	Test plan, code, hypothesis tests, report (2–6)

1. Dataset

Source: Kaggle, "Credit Card Transactions Fraud Detection Dataset" (Kartik Shenoy, CC0), generated with Brandon Harris's Sparkov simulator. One row is one card transaction by one of 999 customers at one of 693 merchants, 1 Jan 2019 to 31 Dec 2020. It comes as `fraudTrain.csv` (1,296,675 rows) and `fraudTest.csv` (555,719 rows; later months), which we combine. The response `is_fraud` marks 9,651 frauds (0.52%), so the classes are heavily imbalanced.

What is raw about it: the timestamp and date of birth are text; customer and merchant locations are four separate coordinates rather than a distance; amounts are extremely skewed (median $47, maximum $28,949); and the file carries names, street addresses and card numbers that identify people.

	Raw file	After encoding
Instances (rows)	1,852,394	1,852,394
Attributes (columns)	23	28
Instances × attributes	42,605,062	51,867,032

Size check: $1{,}852{,}394 \times 28 = 51{,}867{,}032 \geq 3{,}000{,}000$, so the requirement is met. The 28 columns are 13 numeric or derived features, the response, and 14 one-hot category columns.

Key features

Feature	Type	Real-world meaning
`is_fraud`	nominal	transaction was fraudulent (1) or not (0); our response
`amt`	continuous	charge in US dollars
`category`	nominal	merchant type, 14 levels (e.g. `grocery_pos` = in-store groceries, `shopping_net` = online shopping)
`hour`	discrete	hour of the transaction, from the timestamp
`age`	continuous	cardholder's age at the time of purchase, from date of birth
`distance_km`	continuous	distance from cardholder's home to the merchant (derived)
`city_pop`	discrete	population of the cardholder's city
`gender`	nominal	cardholder's gender

2. Data preparation

Issue found	What we did	Why, and not the alternative
Missing values, duplicates	checked all 23 columns and the transaction id: 0 missing, 0 duplicates	nothing to impute or drop, so no rows are lost and no values are invented
Timestamp stored as text	parsed; derived hour, night flag (22:00–03:59), day of week, month	time of day is one of our questions; the raw string cannot be compared
Date of birth, not age	age at transaction $= (\text{date} - \text{dob})/365.25$	age at purchase is what matters; age today would shift every row by up to two years
Four coordinates, no distance	haversine distance between home and merchant, in km	our Test 5 is about distance; no single raw column holds it
`amt` skewness 40.8; 95,054 points above the IQR fence	kept every row; analysed $\log(1 + \texttt{amt})$	the large amounts are where fraud is (median fraud $390 vs $47), so deleting or capping them would remove frauds
`category`, 14 levels	one-hot encoded	nominal, no order; needed for the classifier and the size check
Names, street, card number, transaction id, merchant, job, city, zip, state	dropped	they identify people or are high-cardinality identifiers with no role in our questions

3. What changed after preparation

Measure	Before	After
Rows	1,852,394	1,852,394 (none lost)
Attributes	23	28 (identifiers and index dropped; 5 derived; category one-hot)
Missing values, duplicates	0, 0	0, 0
Skewness of amount	40.81	-0.30 (log scale)
Points beyond the IQR fence	95,054	1,220 (log scale)
Age, distance ranges	not available	13.9–96.2 years; 0.02–152.1 km

The log transform turns a long right tail into a near-symmetric distribution (Figure 1), which is what makes a t-test on amounts reasonable. Deriving age, hour and distance turned three text or coordinate fields into the quantities our questions are about. All derived values are in plausible ranges, so no further cleaning was needed.

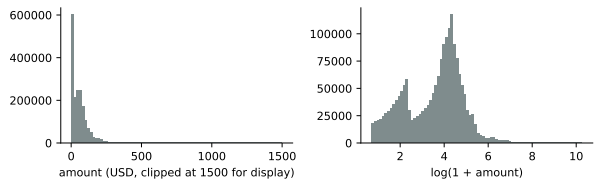


Figure 1: Amount before (left, clipped at $1,500 for display) and after (right) the log transform.

4. Basic questions and problem statement

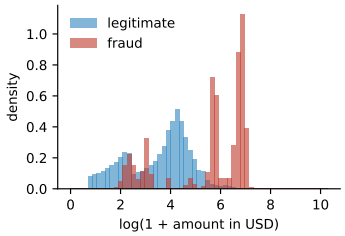


Figure 2: Frauds are far larger: median $390 against $47 for legitimate transactions.

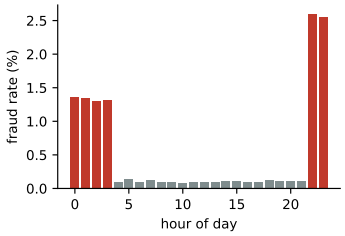


Figure 3: The fraud rate is about 2.6% at 22–23h and 1.3% at 0–3h, against about 0.1% in every other hour (night hours in red).

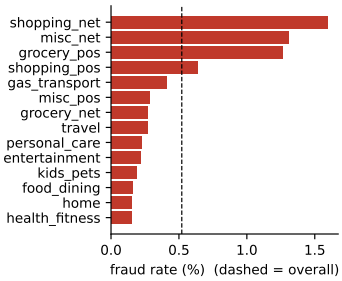


Figure 4: Online shopping, misc. online and in-store groceries have fraud rates of 1.3–1.6%, about 2.4–3× the overall 0.52% (dashed).

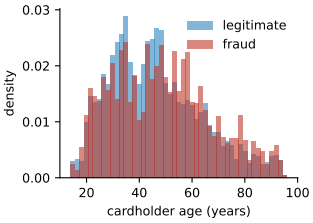


Figure 5: Fraud victims are slightly older (mean 48.8 against 46.3 years), and the shapes differ.

Basic questions

1. How common is fraud, and is it spread evenly over the two years?
2. Which merchant categories see the most fraud?
3. Are fraudulent transactions more expensive?

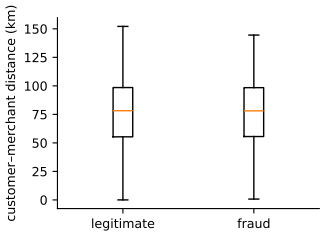


Figure 6: Home-to-merchant distance looks the same for both classes (mean 76.3 against 76.1 km).

Our problem statements

P1 (primary). Do fraudulent transactions differ from legitimate ones in *how much*, *when* and *where* money is spent (amount, time of day, merchant category) by margins large enough to matter to a bank's fraud team?

P2. Are the cardholders hit by fraud different: are they older, and does fraud happen farther from their home?

How they are novel: the basic questions describe fraud rates. Ours fix, before testing, the smallest difference that would matter, test on a power-calculated sample rather than on all 1.85 million rows, use an engineered distance feature, and then check (Section 5.7) whether each tested signal is one a classifier actually needs.

5. Statistical inferences

Rows tested. Not the full data: on 1.85 million rows every test would reject H_0 for differences far too small to matter. We draw a **stratified random sample of 2,000 fraud and 2,000 legitimate transactions** (seed 42, without replacement). A simple random sample would not work: 4,000 random rows hold only about 21 frauds. The size comes from a **power calculation**: a separate pilot sample of 500 per class gave the standard deviations, and $n = 2(z_{1-\alpha} + z_{1-\beta})^2\sigma^2/\Delta^2$ per group at $\alpha = 0.01$ (Bonferroni for five tests) and power 0.8 requires 263 (amount), 383 (night share), 810 (age) and 786 (distance) per group, so 2,000 covers every test. The sample is disjoint from the pilot and representative: within each class, mean amount, age, distance and night share are within 6% of the full data, and no category share differs by more than 1.1 points. Sample ids are saved in the repository.

How we picked each test. For every test we ran all the alternatives in the plan, but the main test was fixed in advance from its assumptions (independent groups, response type, equal or unequal variances, sample size), never from which one gave the smallest p-value; choosing by p-value would bias the result. Part b) of each test reports the alternatives' results; every alternative test of the same hypothesis reaches the same decision as the main test.

Consequence. Fixing 2,000 per class makes the sample's fraud rate (50%) artificial, so Tests 2 and 3 compare how night-time and categories are distributed *within* each class. Independence is symmetric, so this answers the same question, and odds ratios stay valid (the case–control argument). Each test uses $\alpha = 0.05$; we also report whether it survives Bonferroni's 0.01.

Test 1 (P1): are fraudulent transactions larger?

a) Hypotheses, written before running the test

$\boldsymbol{H_0}$	$\mu_F - \mu_L \leq 0$
$\boldsymbol{H_1}$	$\mu_F - \mu_L > 0$
Significance level	$\alpha = 0.05$ (also 0.01)
Side	one-sided (right)
Test	Welch t-test on $\log(1 + \texttt{amt})$
Smallest difference that matters	fraud amounts $1.5\times$ larger ($\Delta = \ln 1.5 = 0.405$)

μ_F, μ_L: mean log amount of fraudulent and legitimate transactions. One-sided, because P1 expects frauds to be larger. On the log scale a difference is a ratio of typical amounts; a fraud team would not change a rule for less than a 50% gap.

b) Other tests we considered

Two independent groups, a continuous response and unknown σ call for a two-sample t-test. An F-test rejects equal variances ($F = 1.66$, df $(1999, 1999)$, $p < 10^{-300}$), so we use Welch's version.

Also considered	Why we did not use it
Two-sample z-test	needs the population σ, which we do not know. Run anyway: $z = 43.67$, same decision
Pooled t-test	assumes equal variances, which the F-test rejects. Run anyway: $t = 43.67$ (3,998), same decision

c) Result

Estimate	$+2.02$ on the log scale: fraud amounts $7.5\times$ larger (means 5.56 vs 3.54)
95% confidence interval	1.93 to 2.11, i.e. $6.9\times$ to $8.2\times$
Test statistic (d.f.)	$t = 43.67$ (3,763.8)
p-value	$< 10^{-300}$
Decision	reject H_0 (also at 0.01)

Frauds are about 7.5 times larger than legitimate transactions. The whole interval is far above the $1.5\times$ that matters, so the difference is real and large.

Test 2 (P1): is fraud concentrated at night?

a) Hypotheses, written before running the test

$\boldsymbol{H_0}$	$p_F - p_L \leq 0$
$\boldsymbol{H_1}$	$p_F - p_L > 0$
Significance level	$\alpha = 0.05$ (also 0.01)
Side	one-sided (right)
Test	two-proportion z-test
Smallest difference that matters	10 percentage points

p_F, p_L: share of fraudulent and of legitimate transactions made at night (22:00–03:59). One-sided, because P1 expects fraud to favour the night. We judged a smaller shift too weak to base a time-of-day rule on.

b) Other tests we considered

Two independent groups and a binary outcome (night or not) call for a two-proportion z-test; with 2,000 per group the normal approximation is safe.

Also considered	Why we did not use it
χ^2 test on the 2×2 table	equivalent ($\chi^2 = 1544.6 = z^2$) but two-sided only
Fisher's exact test	exact, but unnecessary with 2,000 per group; it agrees
Risk ratio	cannot be estimated from a class-stratified sample; we report the odds ratio instead

c) Result

Estimate	$+61.9$ points (85.4% of frauds vs 23.5% of legitimate at night); odds ratio 19.0
95% confidence interval	$+59.5$ to $+64.3$ points; odds ratio 16.2 to 22.4
Test statistic	$z = 39.30$
p-value	$< 10^{-300}$
Decision	reject H_0 (also at 0.01)

Most frauds happen at night, against under a quarter of legitimate spending. The interval is six times the 10 points that matter.

Test 3 (P1): does fraud depend on the merchant category?

a) Hypotheses, written before running the test

$\boldsymbol{H_0}$	category is independent of `is_fraud` (same category mix in both classes)
$\boldsymbol{H_1}$	the category mix differs between the classes
Significance level	$\alpha = 0.05$ (also 0.01)
Side	not applicable (χ^2, upper tail)
Test	χ^2 test of independence, 14×2 table, df $= 13$
Smallest difference that matters	a category with odds ratio ≥ 2 (fraud vs legitimate)

An odds ratio of 2 means a category is twice as over-represented among frauds; below that, flagging a whole category would not be worth the false alarms.

b) Other tests we considered

Two nominal variables call for a χ^2 test of independence; the smallest expected count is 49.5, so the approximation is valid.

Also considered	Why we did not use it
G-test (likelihood ratio)	asymptotically equivalent; run as a check: $G = 778.7$, same decision
Fisher's exact test	impractical for a 14×2 table with 4,000 rows, and unnecessary with large expected counts
Per-category two-proportion tests	14 tests inflate false positives, so not the main test. Run as a post-hoc follow-up with Bonferroni ($\times 14$): 12 of 14 categories differ; only `gas_transport` and `travel` do not

c) Result

Estimate	Cramér's $V = 0.43$ (strong). Odds ratios: `shopping_net` 3.71, `grocery_pos` 3.22, `misc_net` 2.48; lowest `home` 0.18
95% confidence interval	3.05–4.52; 2.67–3.88; 1.96–3.14; 0.13–0.26
Test statistic (d.f.)	$\chi^2 = 742.0$ (13)
p-value	3.6×10^{-150}
Decision	reject H_0 (also at 0.01)

Fraud depends strongly on category. Online shopping and in-store groceries clear the odds-ratio-2 bar with their whole interval; misc. online very nearly does.

Test 4 (P2): are fraud victims older?

a) Hypotheses, written before running the test

$\boldsymbol{H_0}$	$\mu_F - \mu_L = 0$ (mean age)
$\boldsymbol{H_1}$	$\mu_F - \mu_L \neq 0$
Significance level	$\alpha = 0.05$ (also 0.01)
Side	two-sided
Test	Welch t-test
Smallest difference that matters	3 years

Two-sided, because we had no prior direction. We judged a gap under 3 years too small to justify age-specific fraud rules.

b) Other tests we considered

Two independent groups, a continuous response, unknown σ: a two-sample t-test, Welch's version fixed in advance (SDs 18.7 and 18.1).

Also considered	Why we did not use it
Kolmogorov–Smirnov	tests any difference in shape, not the mean; run as a complement: $D = 0.107$, $p = 2 \times 10^{-10}$
TOST equivalence test	used as a follow-up, to ask whether the difference is below 3 years

c) Result

Estimate	$+2.6$ years (49.2 vs 46.7 in the sample)
95% confidence interval	$+1.4$ to $+3.7$ years
Test statistic (d.f.)	$t = 4.40$ (3,994.2)
p-value	1.1×10^{-5}
Decision	reject H_0 (also at 0.01)

Fraud victims are older, but the interval contains the 3 years that would matter, and an equivalence test cannot rule out a 3-year gap either (TOST $p = 0.23$). The difference is real; whether it matters is unresolved. The KS test adds that the age *distributions* differ in shape (Figure 5).

Test 5 (P2): does fraud happen farther from home?

a) Hypotheses, written before running the test

H_0	$\mu_F - \mu_L = 0$ (mean distance)
H_1	$\mu_F - \mu_L \neq 0$
Significance level	$\alpha = 0.05$ (also 0.01)
Side	two-sided
Test	Welch t-test on `distance_km`
Smallest difference that matters	5 km

Two-sided: fraud could be farther (stolen card used elsewhere) or nearer. Under 5 km is within one town, too small for a location rule.

b) Other tests we considered

Again two independent groups and a continuous response with unknown σ, so Welch's t-test (SDs 28.8 and 28.7).

Also considered	Why we did not use it
Kolmogorov–Smirnov	tests shape; $p = 0.064$
TOST equivalence test	used as a follow-up, because failing to reject alone would prove nothing

c) Result

Estimate	$+1.8$ km (76.6 vs 74.8 km)
95% confidence interval	-0.02 to $+3.54$ km
Test statistic (d.f.)	$t = 1.94$ (3,998.0)
p-value	0.053
Decision	fail to reject H_0

No evidence that fraud happens farther from home. Failing to reject is not proof of equality, so we ran TOST with bounds ± 5 km: it rejects non-equivalence ($p = 0.0002$). The sample was powered for 5 km, so any difference is practically negligible.

5.6 Summary and robustness

Test	Estimate (95% CI)	p	Matters?
1 Amount	$7.5\times$ (6.9–8.2)	$< 10^{-300}$	yes
2 Night	$+61.9$ pts (59.5–64.3)	$< 10^{-300}$	yes
3 Category	$V = 0.43$; OR up to 3.7	10^{-150}	yes
4 Age	$+2.6$ yr (1.4–3.7)	10^{-5}	unclear
5 Distance	$+1.8$ km (-0.02–3.5)	0.053	no

All five decisions at $\alpha = 0.01$ are the same on five further independent samples (seeds 1001–1005). As an extra, a χ^2 goodness-of-fit test shows fraud is not spread over the days of the week like legitimate spending ($\chi^2 = 88.5$, df 6, $p = 6 \times 10^{-17}$).

5.7 Using the signals: choosing a classifier

Since the data is labelled, we fit supervised binary classifiers and tried *every* model and class-imbalance option in our plan, then picked the best by a rule fixed in advance.

Set-up. Training rows: `fraudTrain.csv`, split 80/20 into training (1,037,340) and validation (259,335). Test rows: `fraudTest.csv` (555,719 rows, 2,145 frauds; later months), used once, never to choose anything. Features: log amount, hour, night flag, day of week, age, gender, log city population, distance and 14 category dummies. Metric: PR-AUC, plus precision, recall and F1 at a threshold that maximises F1 on validation; not accuracy, since calling everything legitimate already scores 99.6%.

Selection. *Stage 1*, on a stratified 25% subsample of the training split: for each of 11 models, try four imbalance options (none; class weights; random undersampling to 1 fraud per 10 legitimate; SMOTE to the same ratio, on training rows only), keep the best by validation PR-AUC, then try a small hyperparameter grid. *Stage 2*: refit each model's best set-up on the full training split. **Rule: the final model is the one with the best validation PR-AUC.**

What the imbalance options did (Table 3). No single option is best. Class weights help only LightGBM and the random forest and badly hurt the linear models (e.g. spline logistic regression 0.76 $\rightarrow$ 0.31), because they distort predicted probabilities that the threshold already handles. SMOTE helps k-NN (0.59 $\rightarrow$ 0.64) and is level for the MLP. Undersampling is never best except for Naive Bayes, the weakest model.

Result (Tables 1 and 2). By our rule the chosen model is **XGBoost** (validation PR-AUC 0.953; test PR-AUC 0.875, F1 0.82, precision 0.85, recall 0.79). Tree ensembles and the MLP form a top group (test PR-AUC 0.87–0.90); linear models, Naive Bayes and k-NN trail far behind, because the effect of amount is not monotone and depends on the category, which trees capture directly. XGBoost is clearly better than spline logistic regression (McNemar $p \approx 10^{-102}$; PR-AUC $+0.16$ to $+0.19$).

An honest caveat. On the test file **CatBoost** is significantly better than XGBoost (McNemar $p = 10^{-7}$; PR-AUC 0.021–0.030 higher, F1 0.013–0.027 higher). The random forest and LightGBM also make significantly fewer errors than XGBoost at their thresholds, and XGBoost's F1 is not distinguishable from the single decision tree's ($p = 0.78$). Validation ranked the top models differently from the test because our validation rows were a random split of the *same* months as training, while the test file is *later* months: every top model loses 0.03–0.08 PR-AUC from validation to test. We keep XGBoost as the pre-registered choice; CatBoost is the stronger model on unseen months, but its test score is optimistic because we only know it is best after looking at the test file. A validation split by time would have been the better design.

Other options. Adding a per-category fraud-rate feature (target encoding from training rows only) did not help: validation PR-AUC $0.953 \rightarrow 0.952$ for XGBoost and $0.762 \rightarrow 0.762$ for spline logistic regression, so we kept the one-hot categories.

Model	Imbalance option	Val. PR-AUC	Precision	Recall	F1	Test PR-AUC	ROC-AUC
XGBoost	**none**	**0.953**	**0.85**	**0.79**	**0.82**	**0.875**	**0.997**
LightGBM	class wt.	0.952	0.91	0.76	0.83	0.870	0.996
CatBoost	none	0.943	0.88	0.81	0.84	0.901	0.998
Random forest	class wt.	0.931	0.88	0.80	0.84	0.883	0.989
Neural net (MLP)	SMOTE	0.910	0.87	0.79	0.83	0.882	0.998
Decision tree	none	0.903	0.87	0.78	0.82	0.845	0.979
Logistic regr. (splines)	none	0.762	0.64	0.70	0.67	0.703	0.949
k-nearest neighbours	SMOTE	0.692	0.69	0.55	0.61	0.615	0.954
Logistic regr. (linear)	none	0.459	0.32	0.58	0.41	0.391	0.901
Linear SVM	none	0.424	0.63	0.34	0.45	0.364	0.893
Naive Bayes	undersample	0.257	0.46	0.30	0.36	0.205	0.826

Table 1: Stage 2: all 11 models refitted on the full training split, ordered by validation PR-AUC (the selection rule). Precision, recall, F1, PR-AUC and ROC-AUC are on the held-out test file. Bold: the chosen model.

XGBoost against	XGB right only / other right only	McNemar p	F1 difference (95% CI)	PR-AUC difference (95% CI)
LightGBM	122 / 177	0.0018	-0.013 to $+0.004$	-0.000 to $+0.010$
CatBoost	83 / 169	10^{-7}	-0.027 to -0.013	-0.030 to -0.021
Random forest	156 / 237	10^{-4}	-0.028 to -0.010	-0.014 to -0.001
Neural net (MLP)	221 / 253	0.15	-0.015 to $+0.004$	-0.014 to $+0.001$
Decision tree	226 / 233	0.78	-0.009 to $+0.009$	$+0.023$ to $+0.040$
Logistic regr. (splines)	998 / 241	10^{-102}	$+0.140$ to $+0.168$	$+0.160$ to $+0.187$

Table 2: The chosen model against the next five by validation PR-AUC and the logistic-regression baseline, on the test file. McNemar's test (H_0: equal error rates) uses the discordant counts; differences are XGBoost minus the other model, with paired bootstrap intervals (200 resamples).

Model	None	Class wt.	Under.	SMOTE
Logistic regr. (linear)	**0.454**	0.280	0.373	0.332
Logistic regr. (splines)	**0.761**	0.315	0.625	0.570
Naive Bayes	0.230	0.230	**0.251**	0.235
k-nearest neighbours	0.586	–	0.560	**0.636**
Decision tree	**0.848**	0.701	0.724	0.815
Random forest	0.898	**0.906**	0.871	0.893
LightGBM	0.709	**0.921**	0.887	0.917
XGBoost	**0.921**	0.917	0.892	0.913
CatBoost	**0.925**	0.919	0.900	0.912
Linear SVM	**0.421**	0.262	0.371	0.334
Neural net (MLP)	0.881	0.853	0.662	**0.881**

Table 3: Stage 1: validation PR-AUC of each imbalance option (25% training subsample, default settings). Bold: chosen. k-NN cannot take sample weights.

Do the tested signals matter in the model? Dropping each tested feature group and refitting, the fall in test PR-AUC (95% bootstrap CI) is:

Dropped	XGBoost	Spline logistic
Amount (T1)	0.55 (0.53–0.57)	0.48 (0.46–0.50)
Category (T3)	0.27 (0.25–0.29)	0.32 (0.29–0.34)
Time of day (T2)	0.17 (0.16–0.18)	0.11 (0.09–0.13)
Age (T4)	0.07 (0.06–0.08)	0.007 (0.005–0.010)
Distance (T5)	0.00 (-0.004–0.001)	0.00 (0.00–0.00)

The three signals our tests found large and practically important are the ones the models cannot do without; age, significant but small, adds a little (more for XGBoost, which can use it in interactions); distance, shown negligible by the equivalence test, adds nothing. The tests and the classifiers tell the same story.

6. Limitations

- **Simulated data.** The transactions come from the Sparkov generator, so our effects describe its fraud profiles, not a real bank's.
- **Association, not cause.** Night-time or online purchases are not what makes a transaction fraudulent; they are where fraud shows up.
- **Transactions are not fully independent.** 999 customers each make many transactions, so the effective sample is smaller than 4,000 and our p-values are somewhat optimistic.
- **One test sample.** Results come from one stratified sample, though five further samples gave the same decisions.
- **No absolute fraud rates from the sample.** Stratified sampling cannot estimate $P(\text{fraud} \mid \cdot)$; such rates are reported only descriptively from the full data.
- **Model selection.** Our validation rows came from the same months as training, so validation overstated performance on later months and ranked the top models differently from the test file (Section 5.7). A time-ordered validation split would fix this.
- **Multiple tests.** Five tests are run; we report both $\alpha = 0.05$ and Bonferroni's 0.01, and every decision is the same at both.